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Initiation of peri-operative SGLT2 inhibitor reduces mortality and cardiovascular events after surgical aortic valve replacement

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Abstract

Background SGLT2 inhibitors (SGLT2i) demonstrate cardiovascular advantages for patients with heart failure and chronic kidney disease, especially in transcatheter aortic valve replacement. However, no studies have evaluated their perioperative use in surgical aortic valve replacement (SAVR). We examined the association between preoperative initiation of SGLT2i and measured the 1-year postoperative SAVR outcomes. We performed a retrospective analysis using de-identified patient data from the TriNetX Research Network database. Adults who underwent SAVR with first-time SGLT2i use within 1 month preoperatively were compared to a matched group of non-users with no prior history of SGLT2i therapy. Using 1:1 propensity score matching, adjusting for demographics, BMI categories, comorbidities, labs, and medications, we examined primary outcomes of 1-year all-cause mortality and major adverse cardiovascular events (MACE).

Results After propensity score matching (Table 1), our study included 440 patients per group. Baseline characteristics were well balanced post-matching, with a few exceptions.

SGLT2i initiation was associated with significantly lower all-cause mortality at 1 year following SAVR (5.9% vs. 12.3%; OR 0.45, 95% CI 0.28–0.73; p = 0.001) but not with MACE (OR 0.73, 95% CI 0.51–1.03; p = 0.075), cerebral infarction, atrial fibrillation/flutter, or hospital readmission. There were no statistically significant differences in the odds of MI, VT, or SGLT2i-associated complications such as AKI or UTI (Table 2).

Conclusions Preoperative initiation of SGLT2i in SAVR patients was associated with reduced 1-year all-cause mortality.

Keywords Sodium-glucose transporter 2 inhibitors, Surgical aortic valve replacement, Perioperative outcomes, Heart failure, Acute kidney injury

Background

Aortic valvular disease remains one of the most common valvular diseases in Western countries as the population continues to age [1]. Aortic valve disease affects a significant portion of the elderly population, with an estimated 25% of individuals over the age of 65 having some form of degenerative aortic valve disease, and 10% of those over 80 having aortic stenosis [2]. Aortic valve replacement (AVR) is an intervention used to correct aortic valve dysfunction and is a definitive treatment for many aortic valve pathologies. Common conditions treated by AVR include severe aortic stenosis and aortic regurgitation

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(AR). AVR can be performed surgically (SAVR) or via transcatheter techniques (TAVI), and can utilize a variety of different types of valves depending on the patient's need and clinical context [3]. SAVR has an estimated operative mortality of 3% and can include other complications such as pseudoaneurysm, patch dehiscence, infection, hemorrhage, and endocarditis [4, 5]. The presence of atherosclerotic risk factors, such as type 2 diabetes mellitus (T2DM), hyperlipidemia, and obesity, is a clinical predictor of poor outcome for AVR [6].

Sodium-glucose transport protein 2 inhibitors (SGLT2i) are antihyperglycemic drugs that block sodium-glucose transporter proteins, preventing reabsorption of glucose at the renal proximal tubules, resulting in lower blood glucose levels. The subsequent natriuresis produced by SGLT2is reduces the effective circulating volume, producing a secondary antihypertensive effect [7]. Currently, SGLT2i therapy is primarily used as an adjunctive or alternative therapy for treatment of T2DM with significant reductions in HbA_{1c} by 0.37% to 1.16% in patients taking canagliflozin when used as a monotherapy or add-on therapy. Current literature also demonstrates potential multi-organ protective effects of SGLT2i when used in the outpatient setting, including preventing cardiovascular death and heart failure, progression of renal disease, and hepatic fibrosis [8]. SGLT2i use is also generally well tolerated with minimal adverse side effects when used as chronic outpatient therapy, although there is an increased risk of genital infections [9].

Despite the variety of recent clinical trials demonstrating their effectiveness and safety, they remain relatively underutilized. This is especially true when considering their usefulness beyond their antihyperglycemic effect as an organ protective agent for heart failure and kidney disease [10].

However, in the preoperative setting, SGLT2i therapy is often held for a few days prior to surgery, due to concerns for increased risk of euglycemic ketoacidosis (eKA) and acute kidney injury (AKI). A prior multicenter, case control study looked at the postoperative outcomes of those on SGLT2i therapy within 3 months prior to undergoing an inpatient surgical procedure and compared postoperative outcomes to a control group. They found those on SGLT2is had an associated 11% higher risk of eKA, but also approximately 30% reduction in both AKI and 30-day mortality [11]. Prior studies show a potential usage for SGLT2i perioperatively in the setting of cardiac surgery specifically, demonstrating associated cardioprotective and nephroprotective effects in patients with heart failure and chronic kidney disease [12, 13]. Dapagliflozin, an SGLT2i, has previously been demonstrated to have an associated 20% reduction in mortality from an acute heart failure exacerbation event [14]. To the best of our knowledge, no prior studies have specifically investigated the perioperative use of this intervention in patients undergoing SAVR. Given that patients undergoing SAVR often have significant comorbidities, there is growing interest in exploring the potential perioperative benefits of SGLT2i, especially as the prevalence of aortic valvular disease continues to rise. Our study aimed to investigate whether perioperative initiation of SGLT2i therapy has an association with 1-year postoperative SAVR outcomes.

Methods

Data source and study design

This study was a retrospective cohort analysis that utilized de-identified electronic health record (EHR) data from the TriNetX Research Network. TriNetX is a federated database that compiles real-time clinical data, including diagnoses, procedures, medications, laboratory results, and outcomes, from more than 109 healthcare organizations across the United States, representing a population of over 156 million patients. Using this platform, we examined the relationship between perioperative SGLT2i initiation and postoperative outcomes in patients undergoing SAVR. All queries and analyses were performed on July 6, 2025. Since the dataset is entirely de-identified, the study met criteria for exemption from institutional review board approval and informed consent.

Patient selection and exposure groups

The research study included adults aged 18 and above who received SAVR from January 2010 through December 2024 based on CPT procedural codes (33405, 33406, 33410, 33411, 33412). Among these patients, we identified the exposure group based on the usage of SGLT2i initiation prior to surgery. SGLT2i usage is defined by the Anatomical Therapeutic Chemical code (ATC) A10BK. Our exposure group consisted of patients who initiated SGLT2i treatment for the first time within 30 days before their SAVR procedure. Only patients who first received SGLT2i therapy during these defined time frames were included in the analysis, while those with any prior SGLT2i exposure before the defined perioperative window were excluded to minimize confounding from long-term therapy effects. Our retrospective cohort study was conducted in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for cohort studies. Study design, data collection, exposure and outcome definitions, statistical analyses, and reporting were structured to comply with these standards to ensure transparency and reproducibility.

Covariates and matching strategy

To minimize baseline differences and reduce potential confounding variables, we applied 1:1 propensity score matching (PSM) independently for the comparative analysis. The propensity scores were generated from a logistic regression model that analyzed patient demographics together with comorbidities, laboratory results, and medication records. The exposure group was compared to a matched group of non-users, consisting of adults aged 18 and above with no prior SGLT2i exposure. Propensity score distributions before and after matching are shown in Supplementary Fig. S1. Demographic variables included age, sex, race, and ethnicity. Additional clinical comorbidity, relevant laboratory data, as well as medications were also matched (Supplementary Table S1). The nearest-neighbor approach with a caliper of 0.1 pooled standard deviations was used for matching to achieve balanced baseline characteristics. Missingness for key laboratory variables was assessed and incorporated into the analysis. In the SGLT 2 cohort and non-user cohort, key laboratory data availability was observed as follows: hemoglobin (91.4% vs. 93.2%), LDL cholesterol (58.9% vs. 59.1%), B-type natriuretic peptide (33.9% vs. 36.8%), and LVEF (20.2% vs. 19.5%), with no significant differences in missing rates between cohorts. To address missing values, both the availability and the actual lab values were included in propensity score matching. The adequacy of matching was evaluated using standardized mean differences (SMDs), with an SMD < 0.1 considered indicative of acceptable covariate balance between SGLT 2 inhibitor users and non-users after matching.

Outcomes

The primary outcomes of interest were all-cause mortality and major adverse cardiovascular events (MACE) within 1 year postoperatively. MACE is defined as the combination of myocardial infarction, ischemic stroke, and all-cause mortality, consistent with clinical guidelines and standard practices in cardiovascular research [15]. Secondary outcomes included the individual components of MACE (myocardial infarction and ischemic stroke), as well as atrial fibrillation or flutter, ventricular tachycardia, and hospital readmission within 1 year following SAVR. In addition to efficacy endpoints, we assessed safety outcomes commonly associated with SGLT 2i therapy, specifically acute kidney injury (AKI) and urinary tract infections (UTIs). All outcomes were identified using standardized diagnostic and procedural codes to ensure reproducibility across healthcare organizations and consistency within the dataset (Supplementary Table S2).

Statistical analysis

All statistical analyses were conducted using built-in analytic tools within the TriNetX platform, which relies on validated methodologies for clinical research. The analysis used propensity score matching to determine the relationship between SGLT 2i initiation and postoperative events through odds ratio (OR), risk ratio (RR), and 95% confidence interval (CI) calculations for each outcome. A two-sided p-value < 0.05 was considered statistically significant. Covariate balance was confirmed using standardized mean differences before and after matching.

Results

Demographics

Our study looked at adult patients who underwent SAVR and were on SGLT2i therapy within 30 days prior to surgery, compared to a control group that underwent SAVR but did not receive SGLT 2i therapy. Prior to matching, 447 patients who were on SGLT 2i preoperatively and 34,503 patients who were not on SGLT 2i therapy were identified as eligible for the study. After propensity score matching to balance baseline patient characteristics and comorbidities between the two cohorts, 440 patients were included in the study for each cohort.

Demographic and patient characteristics including age, gender, race, BMI, comorbidities, medications and certain laboratory values are summarized in Table 1. Propensity score matching the preoperative SGLT 2i and non-user control groups' patient characteristics and demographics yielded well balanced cohorts with a few exceptions. There were statistically significant differences in some of the laboratory values, with the SGLT 2i users having a lower LVEF (46.5 ± 16.3% vs. 53.3 ± 15.7%, p = 0.0055) and higher hemoglobin (12.9 ± 2.37 g/dL vs. 12.5 ± 2.37 g/dL, p = 0.0081) (Table 1).

Pre-operative initiation of SGLT2i

For primary outcomes, we measured major adverse cardiovascular events (MACE) and 1-year all-cause mortality following the SAVR procedure for preoperative SGLT 2i users vs the non-user control group. Secondary outcomes included myocardial infarction (MI), hospital readmission, cerebral infarction, atrial fibrillation and flutter, ventricular tachycardia, urinary tract infection (UTI), and AKI. All outcomes were measured at 1 year. The total patients included were 880 patients, with 440 within each cohort (SGLT2i users and non-users). The results listed below are after propensity score matching between the two cohorts (Table 2).

Preoperative SLGT2i initiation 1 year following SAVR was associated with a statistically significant lower incidence of all-cause mortality with 5.91% of patients dying

Table 1 Baseline demographic, clinical, laboratory, and medication characteristics of the matched groups: preoperative SGLT2 inhibitor users vs matched non-users

	Pre-op SGLT2 users N = 440	Matched pre-op non-users N = 440	Pre-op p-value	Standardized mean **difference**
Demographic				
Age at index (mean ± SD)	64.66 ± 10.26	65.23 ± 10.66	0.42	0.0541
Male	286 (65%)	288 (65.46%)	0.89	0.0095
Female	109 (24.77%)	106 (24.09%)	0.81	0.0159
Unknown gender	45 (10.23%)	46 (10.46%)	0.91	0.0075
Race				
White	296 (67.27%)	285 (64.77%)	0.43	0.0528
Black or African American	40 (9.09%)	47 (10.68%)	0.43	0.0533
Asian	11 (2.50%)	20 (4.55%)	0.10	0.1111
American Indian or Alaska Native	10 (2.27%)	10 (2.27%)	1.00	0
Native Hawaiian or other PI	10 (2.27%)	10 (2.27%)	1.00	0
Other race	15 (3.41%)	12 (2.73%)	0.56	0.0395
Unknown race	74 (16.82%)	70 (15.91%)	0.72	0.0246
Ethnicity				
Hispanic or Latino	26 (5.91%)	20 (4.55%)	0.36	0.0613
Not Hispanic or Latino	311 (70.68%)	323 (73.41%)	0.37	0.0608
Unknown ethnicity	103 (23.41%)	97 (22.05%)	0.63	0.0325
Comorbidities				
Hypertensive diseases	366 (83.18%)	378 (85.91%)	0.26	0.0755
Chronic ischemic heart disease	346 (78.64%)	358 (81.36%)	0.31	0.0682
Heart failure	312 (70.91%)	323 (73.41%)	0.41	0.0558
Diabetes mellitus	273 (62.05%)	289 (65.68%)	0.26	0.0757
Chronic kidney disease	125 (28.41%)	133 (30.23%)	0.55	0.0399
Nicotine dependence	120 (27.27%)	119 (27.05%)	0.94	0.0051
Body Mass Index (BMI)				
BMI ≤ 19.9	10 (2.27%)	10 (2.27%)	1.00	0
BMI 20–29	35 (7.96%)	29 (6.59%)	0.44	0.0525
BMI 30–39	106 (24.09%)	109 (24.77%)	0.81	0.0159
BMI ≥ 40	46 (10.46%)	50 (11.36%)	0.67	0.0292
Laboratory values				
Hemoglobin (g/dL, mean ± SD)	12.95 ± 2.37	12.5 ± 2.37	0.01	1.070266
LDL cholesterol (mg/dL, mean ± SD)	76.3 ± 37.54	82.39 ± 36.03	0.06	0.170431
BNP (pg/mL, mean ± SD)	1405.63 ± 3309.96	1596.6 ± 4378.31	0.67	0.04921
LVEF (% mean ± SD)	46.47 ± 16.34	53.29 ± 15.68	0.01	0.425494
Medications				
Beta blockers	366 (83.18%)	374 (85%)	0.46	0.0497
Statins	371 (84.32%)	386 (87.73%)	0.14	0.0984
Diuretics	353 (80.23%)	367 (83.41%)	0.22	0.0826
ACE inhibitors	187 (42.50%)	184 (41.82%)	0.84	0.0138
ARBs, plain	229 (52.05%)	235 (53.41%)	0.69	0.0273
ARBs, combination	84 (19.09%)	66 (15%)	0.11	0.109
GLP-1 agonists	50 (11.36%)	50 (11.36%)	1.00	0
Insulin	229 (52.05%)	237 (53.86%)	0.59	0.0364

Table 2 Comparative outcomes between SGLT2 inhibitor users and matched non-users in preoperative SAVR cohort

Outcome	Pre-op SGLT2 users (N = 440)	Matched non-users (N = 440)	Odds ratio (95% CI)	Risk ratio (95% CI)	p-value
All-cause mortality	26 (5.91%)	54 (12.27%)	0.45 (0.28–0.73)	0.48 (0.31–0.75)	0.001
Myocardial infarction	46 (10.46%)	53 (12.05%)	0.85 (0.56–1.30)	0.87 (0.60–1.26)	0.46
MACE	66 (15.00%)	86 (19.55%)	0.73 (0.51–1.03)	0.77 (0.57–1.03)	0.075
Cerebral infarct	24 (5.46%)	38 (8.64%)	0.61 (0.39–1.04)	0.63 (0.39–1.04)	0.065
Atrial fibrillation/flutter	194 (44.09%)	216 (49.09%)	0.82 (0.63–1.07)	0.90 (0.78–1.04)	0.14
Ventricular tachycardia	31 (7.05%)	28 (6.36%)	1.12 (0.66–1.89)	1.11 (0.68–1.81)	0.69
Hospital readmission	231 (52.50%)	221 (50.23%)	1.10 (0.84–1.43)	1.05 (0.92–1.19)	0.50
UTI	28 (6.36%)	29 (6.59%)	0.96 (0.56–1.65)	0.97 (0.58–1.60)	0.89
AKI	115 (26.14%)	108 (24.55%)	1.09 (0.80–1.47)	1.07 (0.85–1.34)	0.59

in the SGLT2i group versus 12.27% in the non-user control group (OR 0.45, 95% CI 0.28–0.73; p = 0.001). Major adverse cardiovascular events (MACE) occurred slightly less frequently among SGLT2i users (15.0% vs. 19.6%) but did not show a statistically significant difference compared to the non-user control group (OR 0.73, 95% CI 0.51–1.03; p = 0.075). Kaplan–Meier analysis was performed for all-cause mortality (Fig. 1).

For secondary outcomes, there were no detectable statistically significant differences between the two groups for any of the measured secondary outcomes at 1 year. Observed rates for myocardial infarction showed no detectable statistical difference (10.5% vs. 12.1%; OR 0.85, 95% CI 0.56–1.30; p = 0.46). Similarly, cerebral infarction demonstrated a non-significant reduction in SGLT2 inhibitor users (5.5% vs. 8.6%; OR 0.61, 95% CI 0.39–1.04; p = 0.065). Atrial fibrillation/flutter was also lower in the SGLT2i group (44.1% vs. 49.1%) but no detectable statistical difference (OR 0.82, 95% CI 0.63–1.07; p = 0.14). There were no significant differences in ventricular tachycardia (7.1% vs. 6.4%; p = 0.69) or hospital readmission (52.5% vs. 50.2%; p = 0.50).

SGLT2i therapy has previously been associated with increased rates of UTI and AKI, but in our study, there was no detectable statistical difference between the two groups for urinary tract infection (6.4% vs. 6.6%; OR 0.96, 95% CI 0.56–1.65; p = 0.89), or acute kidney injury (26.1% vs. 24.6%; OR 1.09, 95% CI 0.80–1.47; p = 0.59).

Discussion

Our retrospective study of adult patients who received SAVR showed a potential mortality benefit associated with preoperative SGLT2 inhibitor use in patients undergoing SAVR. This is despite the fact that no significant difference is noted in other cardiovascular outcomes including myocardial infarction, MACE, cerebral infarct, atrial fibrillation/flutter, or ventricular tachycardia. We also observed no significant difference in safety metrics for SGLT2i therapy during the 1-year postoperative period, including hospital readmission, urinary tract infection, or acute kidney injury. Our findings suggest that the mortality benefit may be mediated through mechanisms distinct from those traditionally associated with a reduction in discrete cardiovascular events.

In our retrospective cohort study, we observed a mortality benefit associated with preoperative SGLT2 inhibitor use in patients undergoing SAVR, despite the absence of statistically significant differences in secondary cardiovascular outcomes including myocardial infarction, MACE, cerebral infarct, atrial fibrillation/flutter, ventricular tachycardia, hospital readmission, urinary tract infection, and acute kidney injury. The observed mortality reduction without corresponding decreases in MI or MACE is consistent with emerging evidence that SGLT2 inhibitors exert their cardiovascular protective effects primarily through pathways other than atherothrombotic event prevention. Large-scale meta-analyses have demonstrated that SGLT2 inhibitors reduce MACE primarily by decreasing cardiovascular death rather than by preventing myocardial infarction or stroke [16]. In the SGLT2i Meta-analysis Cardio-Renal Trialists Consortium analysis of 78,607 patients, SGLT2 inhibitors reduced cardiovascular death by 14% (HR 0.86) but showed no significant effect on myocardial infarction (HR 0.95) or stroke (HR 0.99). This pattern aligns with our findings and suggests that the mortality benefit in SAVR patients may stem from improved hemodynamic stability, reduced heart failure decompensation, and prevention of sudden cardiac death, rather than a reduction in ischemic events.

Several potential mechanisms may explain the mortality benefit observed in our study. SGLT2 inhibitors attenuate ischemia–reperfusion injury through reduction of cardiomyocyte necrosis and oxidative stress, which are central to perioperative myocardial damage. These agents improve myocardial energy metabolism by enhancing

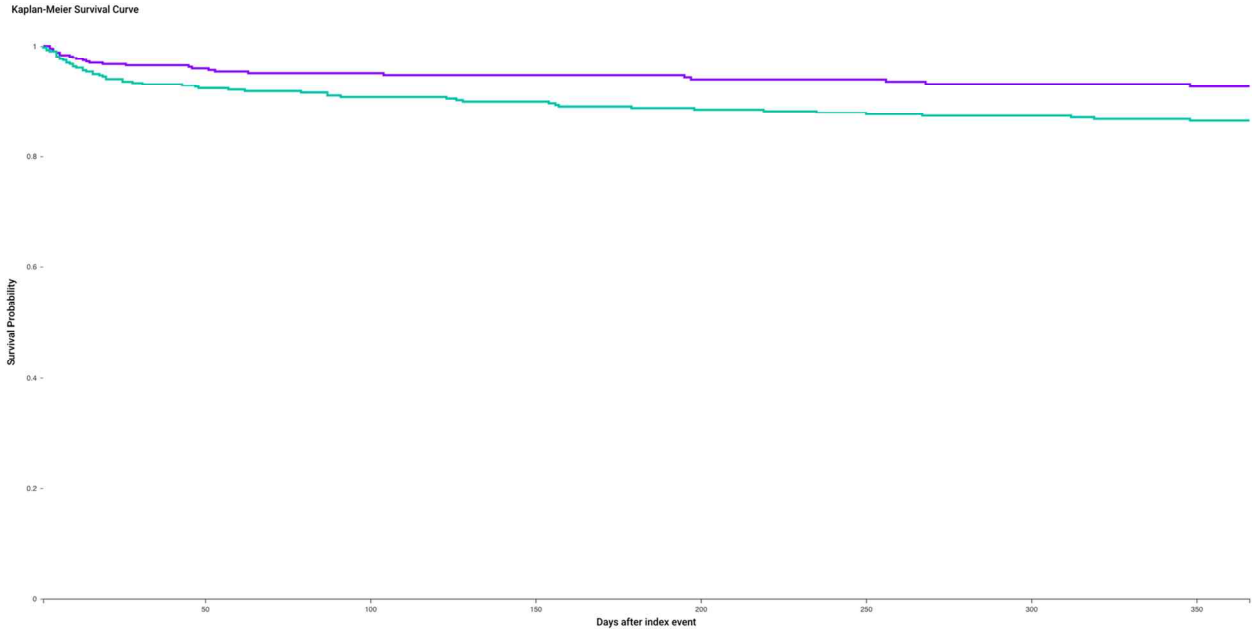


Fig. 1 Kaplan–Meier curve for all-cause mortality

mitochondrial function and increasing ketone utilization, thereby preserving cardiac contractility during the metabolic demands of cardiopulmonary bypass [17, 18]. The reduction in intracellular sodium accumulation in cardiomyocytes improves calcium handling and may reduce the risk of arrhythmias and contractile dysfunction in the perioperative period [3]. Additionally, SGLT2 inhibitors provide preload and afterload reduction through osmotic diuresis and natriuresis, which may optimize volume status entering surgery and reduce cardiac workload postoperatively [19]. A recent randomized pilot study demonstrated that perioperative empagliflozin increased diuresis by 549 mL and reduced positive fluid balance by 1217 mL in the first 24 postoperative hours without increasing vasopressor requirements [20].

The absence of a statistically significant effect on atrial fibrillation/flutter in our study contrasts with findings from large, randomized trials in medical populations. In the DECLARE-TIMI 58 trial, dapagliflozin reduced AF/AFL events by 19% (HR 0.81) [21]. However, the perioperative setting of SAVR presents unique considerations. Postoperative atrial fibrillation following cardiac surgery is multifactorial, driven by acute inflammatory responses, atrial manipulation, pericardial inflammation, and autonomic dysregulation that may overwhelm the antiarrhythmic effects of SGLT2 inhibitors observed in chronic medical conditions. The relatively short perioperative observation period in our study may have been insufficient to capture the longer-term antiarrhythmic benefits that emerge over months to years of SGLT2 inhibitor therapy. Given the limitations of this study, including limited granular data and a modest sample size, the lack of observed differences may reflect insufficient statistical power rather than a true absence of association.

Importantly, in our study findings, preoperative SGLT2i initiation was not associated with increased risk of AKI, UTI, or hospital readmission. Despite prior concerns regarding the risks of AKI, UTIs, and perioperative hemodynamic instability, we found no increase in these complications among users [22]. While many publications still show a conflicting effect of SGLT2i on AKI rates, most recent studies have shown no increased risk of AKI and some even showed a reduction of AKI risk [23]. It was speculated that SGLT2is achieve renal benefits through their ability to stabilize blood pressure and promote sodium excretion and control glomerular filtration pressure, which protects the kidneys during surgery in the perioperative setting [24]. Previous studies have also shown that there is no significant increase in UTI risk associated with SGLT2i use, consistent with our results [25, 26]. However, it is possible that the risk of AKI, UTI, or hospital readmission was not captured in our study due to only a modest study population size.

Baseline laboratory and echocardiographic characteristics are summarized in Table 1. BNP was well balanced between SGLT2i users and non-users (1405.6 ± 3310 vs. 1596.6 ± 4378 pg/mL; SMD 0.049), reflecting the underlying heart failure population. Hemoglobin (12.95 ± 2.37

vs. 12.50 ± 2.37 g/dL; SMD 1.07) and LDL cholesterol (76.3 ± 37.5 vs. 82.4 ± 36.0 mg/dL; SMD 0.17) showed some residual imbalance, though the difference in hemoglobin is unlikely to be clinically meaningful. Left ventricular ejection fraction (LVEF) was lower in the SGLT2i cohort (46.5 ± 16.3% vs. 53.3 ± 15.7%; SMD 0.425), likely reflecting that these patients had more advanced cardiac dysfunction, which may have prompted initiation of SGLT2i therapy. Importantly, given that the SGLT2i cohort experienced either similar or improved outcomes compared with non-users, these baseline differences are unlikely to have materially influenced the observed results.

Strength and limitations

This study has several notable strengths. First, our study leverages a large, multicenter dataset from the TriNetX Research Network, enhancing the generalizability of findings to real-world clinical practice. We also used the built-in rigorous 1:1 propensity score matching functionality on the TriNetX platform to minimize confounding variables by balancing key demographic, clinical, and laboratory variables. This helped us strengthen the validity of the observed associations. Second, to our knowledge, this is the first study to evaluate the impact of perioperative SGLT2 inhibitor initiation specifically in SAVR patients, addressing an important evidence gap. Previously, most of the studies were focused on TAVI as it is gaining more popularity among physicians and patients due to the less-invasive nature of the procedure.

However, several limitations should also be acknowledged. As this was a retrospective cohort study using de-identified registry data, and despite rigorous propensity score matching, the possibility of residual confounding cannot be excluded. As information on medication adherence, precise timing of initiation, and dosage adjustments were not collected by the registry and not available to us, it was impossible for us to adjust and accommodate for this potential confounding effect. Notably, a previous study had shown that only 36% of patients are fully adherent to the SGLT2i medication and dosage schedule in a year [27], which may ultimately influence our study results and treatment effects. Additionally, modest baseline differences in cardiac function, particularly LVEF, could have contributed to observed outcome differences between groups. Although, prior research has shown SGLT2i therapy has the potential to improve functional capacity and quality of life in patients across a large spectrum of LVEF [28]. Higher LVEF pretreatment, however, can lead to more subtle improvements in outcome that may be erroneously attributed to the prescribed SGLT2i treatment. Detailed operative variables, including urgency (elective vs. urgent/emergent), concomitant CABG, valve type/mechanism, and cross-clamp or cardiopulmonary bypass times, were not available in the TriNetX platform. As a result, unmeasured differences in surgical complexity may also have acted as potential confounders.

A further limitation of our study is that the TriNetX platform does not currently support formal competing-risk analyses (e.g., Fine-Gray subdistribution hazard models). As a result, patients who died before experiencing non-fatal outcomes were censored at the time of death, which does not fully account for the competing risk of mortality. This approach may overestimate the cumulative incidence of non-fatal events, particularly in post-SAVR populations with substantial mortality risk, and should be considered when interpreting our findings.

Conclusions

Taken together, these findings suggest a significant association between preoperative initiation of SGLT2i and reduced mortality in patients undergoing SAVR. However, preoperative use was not associated with significant differences in major adverse cardiovascular events (MACE), cerebral infarction, atrial fibrillation/flutter, or hospital readmission. Importantly, preoperative SGLT2 inhibitor initiation was not associated with increased risks of myocardial infarction, ventricular tachycardia, acute kidney injury, or urinary tract infection. These findings support further prospective investigation to confirm these associations and the potential role of preoperative SGLT2 inhibitors in patients undergoing SAVR.

Abbreviations

AKI	Acute kidney injury
AR	Aortic regurgitation
AS	Aortic stenosis
ATC	Anatomical therapeutic chemical
AVR	Aortic valve replacement
BMI	Body mass index
CI	Confidence interval
CPT	Current Procedural Terminology
EHR	Electronic health record
HbA1c	Hemoglobin A1c
HF	Heart failure
IRB	Institutional Review Board
LVEF	Left ventricular ejection fraction
MACE	Major adverse cardiovascular events
MI	Myocardial infarction
OR	Odds ratio
RR	Risk ratio
PSM	Propensity score matching
SAVR	Surgical aortic valve replacement
SGLT2	Sodium-glucose transport protein 2
SGLT2i	Sodium-glucose transport protein 2 inhibitor
T2DM	Type 2 diabetes mellitus
TAVI	Transcatheter aortic valve implantation
UTI	Urinary tract infection
VT	Ventricular tachycardia

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s43057-026-00193-5.

Supplementary Material 1. Supplementary Table S1: Some of the matched variables and their corresponding coding systems and codes. Supplementary Table S2: Diagnostic and procedural codes used to define study outcomes.

Supplementary Material 2. Supplementary Figure S1: Propensity score distributions and overlap before and after matching.

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Authors' contributions

JY contributed to the conception and design of the study, data acquisition, data analysis, interpretation of data, and drafting of the manuscript. JM contributed to data acquisition and drafting of the manuscript. TAJ and NK contributed to study design, data analysis, interpretation of data, and critical revision of the manuscript. RK was responsible for study conception and design, data analysis and interpretation, critical revision of the manuscript, and supervision of the project. All authors read and approved the final manuscript.

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Data availability

The data that support the findings of this study were obtained from the TriNetX Research Network. Restrictions apply to the availability of these data, which were used under license for the current study and are not publicly available. Data are, however, available from the corresponding author upon reasonable request and with permission of TriNetX.

Declarations

Ethics approval and consent to participate

This study was conducted using de-identified data obtained from the TriNetX Research Network. The TriNetX platform provides only anonymized patient information and contains no protected health information. As a result, this study met criteria for exemption from institutional review board approval and informed consent in accordance with applicable regulatory guidelines.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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